Diploma

UNIT –I: Atomic Structure and Chemical Bonding

Basic concepts of atomic structure, Matter wave concept, de-Broglie equation, Heisenberg Uncertainty Principle, Pauli Exclusion Principle, Quantum number, Aufbau Principle, Electronic configuration, Shapes of orbitals, Ionic bond, Covalent bond, Co-ordinate bond, Hydrogen bonding, Valence Bond Theory, Molecular Orbital Theory, Valence Shell Electron Pair Repulsion Theory, Hybridisation.

•Materials → Molecules →Atoms

•Atoms = protons (p) + neutrons (n) + electrons (e)

•Protons and neutrons are made of quarks

	Mass (kg)	Charge (coulomb)
Electron	9.109×10^{-31}	-1.602×10^{-19}
Proton	1.673×10^{-27}	$+ 1.602 \times 10^{-19}$
Neutron	1.675×10^{-27}	0

The Structure of Atoms

Atomic Number (*Z*)

- Moseley introduced the concept of atomic number.
- Atomic number denotes the number of protons in the nucleus. The number of protons in a neutral atom equals the number of electrons.

  In neutral atom:
  No. of protons = No. of electrons = atomic number Z
- Value of Z is different for each element H (Z = 1), O (Z = 8), Fe (Z = 26) etc.

Mass Number (*A*)

- The mass number is the sum of the number of protons and neutrons in the nucleus.

  Atomic mass, A = Z + n

• Atomic weight = weighted average of all isotopes

- An atom with an atomic number and a mass number is denoted by $_{Z}X^{A}$
- Protons and neutrons are referred to collectively as nucleons.
  - **Isotopes:** Same element (same Z) might have different atomic mass i.e. different number of neutrons (n), same chemistry (same number of electron). Hydrogen: Protium ($_1H^1$), Deutirium ($_1H^2$) or D, Tritium ($_1H^3$) or T

- **Isobars:** Isobars are atoms of different chemical elements that have the same number of nucleons. Correspondingly, isobars differ in atomic number but have the same mass number. An example of a series of isobars is $_{16}S^{40}$, $_{18}Ar^{40}$, $_{19}K^{40}$, and $_{20}Ca^{40}$.
- **Isotones:** Isotones are atoms that have the same neutron number but different proton number. For example, $_{16}S^{36}$, $_{17}Cl^{37}$, $_{18}Ar^{38}$, $_{19}K^{39}$, and $_{20}Ca^{40}$ are all isotones of 20 since they all contain 20 neutrons.

- By 1900, it had been discovered that the atom is not a simple, indivisible particle, but rather a collection of sub-atomic particles.
- J.J. Thomson was the first to propose a model for atom structure.

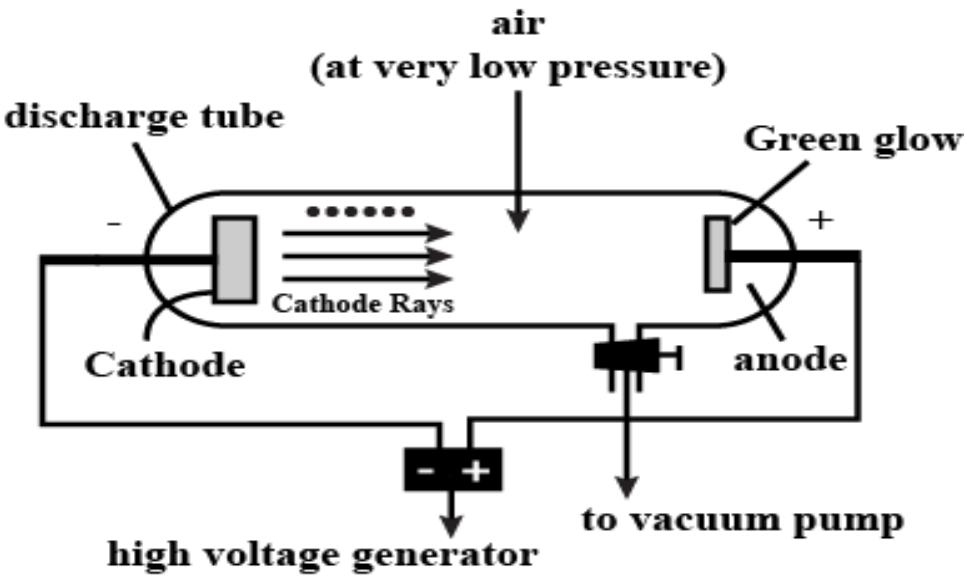
Constituent of Atom: Several phenomena, such as radioactivity, demonstrate that atoms are divisible and consist of three fundamental particles: electron, proton, and neutron.

Electron ($_{-1}e^{0}$)

J.J. Thomson discovered the electron in 1897. When a high-voltage electric charge is passed through a low-pressure gas, a stream of rays is emitted from the cathode surface. These are referred to as cathode rays.

The magnitude of charge on electron is -1.602×10^{-19} C

The mass of electron is 9.109×10^{-31} kg



Properties of Cathode Rays:

- Cathode rays always travel away from the cathode in a straight line, casting shadows on metallic objects in their path.
- Cathode rays cause mechanical motion in the path of a spin wheel. As a result, they have kinetic energy and must be material particles.

- Because the cathode ray beam is deflected towards the positively charged plate, the particles in the cathode rays have a negative charge.
- Cathode rays can be deflected by both an electric and a magnetic field.
- Cathode rays have the ability to penetrate through thin layers of matter.
- Cathode rays emit X-rays when they strike a metallic target with a high melting point, such as tungsten.
- Cathode rays glow when they strike glass or certain other materials.
- The nature of cathode rays is unaffected by the cathode's nature and the gas in the discharge tube.

Charge/Mass (e/m) Ratio

J.J. Thomson determined the charge-to-mass ratio of the electron by measuring the deflection under the influence of both electric and magnetic fields at the same time.

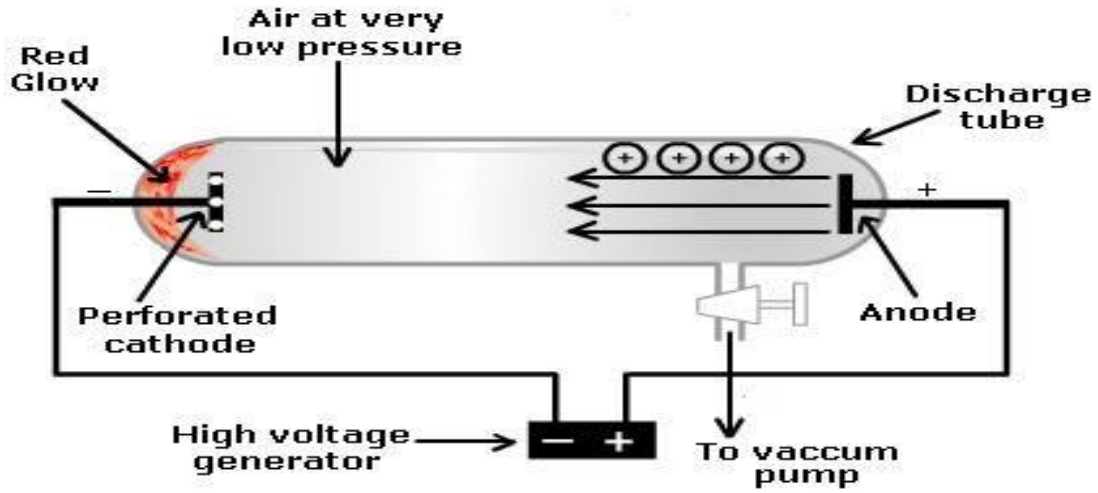
For electrons, the value of the **e/m** ratio was found to – 1.7588 x 10^{11} C kg^{-1}. R.A. Millikan measured the charge on the electron.

Proton ($_1p^1$)

E. Goldstein demonstrated the existence of positively charged particles in an atom in 1886. When a high voltage is applied across the electrodes, a new type of ray is produced that passes through the perforated cathode and causes a glow on the opposite wall to the anode. Anode rays and canal rays are other names for these rays.

The magnitude of charge on proton is + 1.602 $\times 10^{-19}$ C.

The mass of proton is 1.673 $\times 10^{-27}$ kg.



Properties of Anode Rays:

- Anode rays travel in a straight line and cast shadows.
- Anode rays are bent in the opposite direction as cathode rays by the magnetic and electric fields.

- The anode rays can also rotate the wheel in their path and cause heating.
- The charge to mass ratio (e/m) of positive rays is less than e/m of electrons and depends on the nature of the gas used in the tube.

$$\frac{e}{m} = 9.579 \times 10^{7}\ C\ kg^{-1}$$

- When electrons are removed from neutral atoms or molecules of gas, positively charged ions are formed. Positively charged ions move towards the perforated cathode, forming a beam of positive rays.
- Rutherford has given the name proton to anode rays. So the discovery of the proton is credited to Ernest Rutherford.

Neutron ($_0n^1$)

The neutrons were discovered by James Chadwick by bombardment of alpha particle on beryllium.

$$_4Be^9 + {}_2He^4 \rightarrow {}_6C^{12} + {}_0n^1$$

Beryllium alpha particle Carbon Neutron

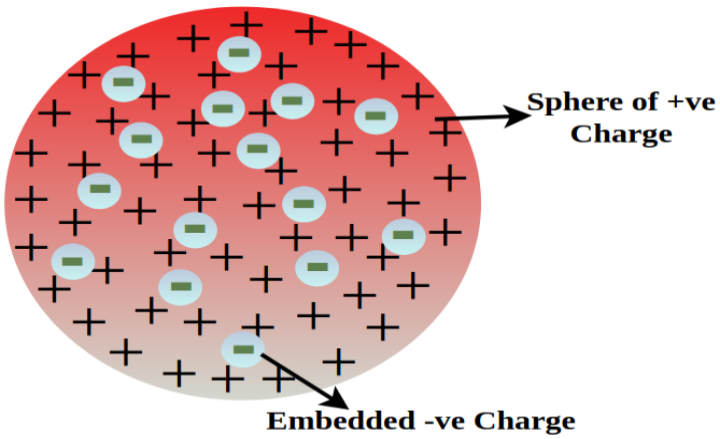
These particles are neutral having charge equals zero. Their mass is approximate same as that of a hydrogen atom, i.e. 1.675×10^{-27} kg.

Atomic Models

Various atomic models were proposed to show the arrangement and distribution of particles (electrons, protons and neutrons) within an atom.

Thomson's Atomic Model:

The first simple model of an atom was proposed by J.J. Thomson. According to Thomson, an atom is a positively charged uniform sphere of radius in which electrons are embedded in such a way that negative charge equals to positive charge. This model is also called plum-pudding model and watermelon model. This model was unable to explain the stability of an atom.



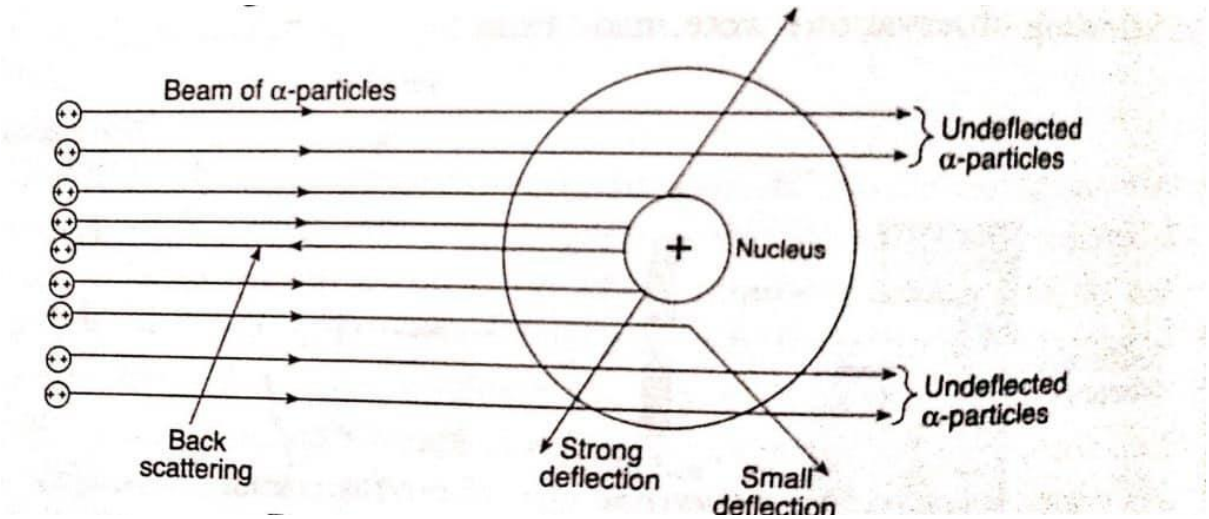
Thomson's Atomic Model

Rutherford's Atomic Model (1912):

Rutherford conducted an experiment by bombarding a thin sheet of gold foil with α-particles and then studied the trajectory of these particles after their interaction with the gold foil.. Rutherford's -particle scattering experiments are concerned with the discovery of the nucleus.

These experiments resulted in the following conclusions:

- A major fraction of the α-particles bombarded towards the gold sheet passed through the sheet without any deflection, and hence most of the space in an atom is empty.
- Some of the α-particles were deflected by the gold sheet by very small angles, and hence the positive charge in an atom is not uniformly distributed. The positive charge in an atom is concentrated in a very small volume.
- Very few of the α-particles were deflected back, that is only a few α-particles had nearly 180° angle of deflection. So the volume occupied by the positively charged particles in an atom is very small as compared to the total volume of an atom.

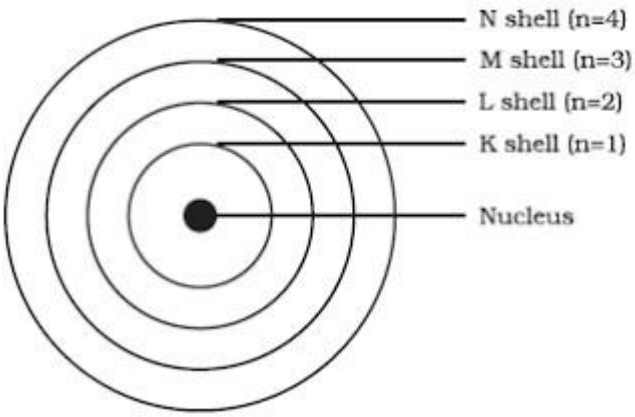


The postulates of the Rutherford model of atom are:

- An atom is mostly made up of empty space. At its core, each atom has a heavy positively charged body known as the nucleus. Most of the mass is concentrated in the nucleus of atoms.
- Planetary electrons are negatively charged electrons that revolve around the nucleus in a circular path.
- The nucleus ($r = 10^{-15}$ m) is very small in comparison to the atom ($r = 10^{-10}$ m).
- A strong interaction force holds all protons and neutrons together in the nucleus.
- This atom model is known as the planetary model of the atom because the sun at the center is compared to the nucleus, planets revolving around the sun are compared to electrons.
- Rutherford's atomic model failed because he was unable to explain the atom's electronic structure. In other words, it does not provide information about the distribution of electrons around the nucleus.

Note: During alpha -particle scattering experiment, particles were bombarded by a thin sheet of heavy metals such as gold and platinum because their nuclei are large, resulting in good results.

Bohr's Atomic Model (1913):



- This atomic model, proposed by Neils Bohr, is based on Planck's quantum theory of radiations.
- Electrons in an atom only revolve around the nucleus in specific circular paths known as orbits. Each orbit contains a fixed amount of energy.
- Only those orbits are permitted in which the electron's angular momentum is a whole number multiple of $\frac{h}{2\pi}$ ('h' is a Planck's constant) i.e.

$$mvr = n\frac{h}{2\pi} \quad \text{(where n = 1, 2, 3.....)}$$

- When an electron jumps from a higher energy level to a lower energy level, energy is emitted, and energy is absorbed when an electron jumps from a lower energy level to a higher energy level.
- When a transition between two different energy levels or states occurs, the frequency of radiation absorbed or emitted is given by

$$\nu = \frac{\Delta E}{h} = \frac{E2-E1}{h}$$

Here, E_1 and E_2 are the energies of lower and higher energy states.

- The energy of an electron in a specific energy level (n) is given by:

$$E_n = -13.6 \times \frac{z2}{n2} \text{ eV/atom}$$

Where Z is atomic number.

Z = 1 for hydrogen atom then energy is:

$$E_n = \frac{-13.6}{n2} \text{ eV/atom}$$

- Bohr's Radius (r) is given by:

$$r = 0.53 \times \frac{n2}{z} \ A^o$$

if Z = 1 then radius is:

$$r = 0.53 \times n^2 \ A^o$$

Limitations of the Bohr's Atomic Model

Bohr's atomic model explains many features of the atomic theory, but it has its limitations. The problems with the Bohr model are listed below:

- It contradicts Heisenberg's Uncertainty Principle. The consideration of the Bohr's atomic model that electrons have both a known orbit and radius violates Heisenberg's Uncertainty Principle
- It fails to predict the spectra of larger atoms and provides insufficient data for the spectra of smaller atoms
- It fails to explain the Zeeman Effect. Zeeman Effect is the splitting of spectral lines into several components in the presence of a magnetic field
- It does not explain the Stark Effect. The Stark effect is the splitting of spectral lines into various components in the presence of an electric field. The Bohr's model of an atom fails to clarify this effect
- It fails to explain the shapes of molecules.

Bohr's model can explain why electrons do not fall into the nucleus. Niels Bohr postulated that electrons revolve in fixed energy levels. The energy levels have discrete energies associated with them. Bohr's model was the first one to incorporate quantum theory.

The model was later modified by Sommerfeld due to its limitations. Sommerfeld's model assumed the orbits of electrons to be elliptical rather than circular. Sommerfeld's model could explain spectral effects, but it had limitations too.

Quantum Mechanical Model of the Atom

The formulation of a quantum mechanical model of an atom is based on two key concepts.

1. Dual Nature of Material Objects (de-Broglie Concept)

He stated that every matter or particle has some sort of wavelength associated with it when it is in motion. In 1924 de-Broglie proposed that matter such as radiation acts

as both particle and wave. He derived the relation between momentum and wavelength of light. Einstein suggested that the light is made up of small packets called quanta or photons. Hence if light is a particle, it must have energy associated with it, given by:

$E = mc^2$

And if light is wave, its energy is given by Planck's theory, given by:

$E = \frac{hc}{\lambda}$

As both energy are the same, hence equating:

$$mc^2 = \frac{hc}{\lambda}$$

$$\lambda = \frac{h}{mc}$$

According to de-Broglie, the wavelength associated with a particle of mass (m), moving with velocity (v) is given by

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

here, h = Planck's constant, p = momentum .This equation is known as de-Broglie equation.

2. Heisenberg's Uncertainty Principle:

In 1927, Heisenberg pointed out that measuring the position and momentum (or velocity) of a microscopic particle at the same time is impossible with absolute precision. If Δx is the error in position measurement and Δp is the error in the measurement of momentum, then

$$\Delta x \text{ X } \Delta p \geq \frac{h}{4\pi}$$

Since momentum, p = mv, Heisenberg's uncertainty principle formula can be alternatively written as,

$$\Delta x \text{ X } m.\Delta v \geq \frac{h}{4\pi}$$

Where, ΔV is the error in the measurement of velocity, and assuming mass remains constant during the experiment,

$$\Delta x \text{ X } \Delta v \geq \frac{h}{4\pi\, m}$$

Accurate measurement of position or momentum automatically indicates larger uncertainty (error) in the measurement of the other quantity.

Quantum Numbers

Four sets of quantum numbers completely describe the position and nature of an electron.

1. Principal Quantum Number (n): (Bohr)

It is denoted by n. It refers to the electron shell with the most electrons. It gives the electron's likely distance from the nucleus. The bigger the value of 'n,' the greater the distance between the electron and the nucleus.

It also indicates the size of the electron's orbital and energy level. As a result, a bigger 'n' signifies a larger orbital size and thus a larger atomic radius.

n = 1, 2, 3, …, (only positive integers). The various shells are denoted as K, L, M, … etc.

2. Azimuthal Quantum Number (l): (Sommerfeld)

The azimuthal (or the orbital angular momentum) quantum number is defined as the determination of the shape of an orbital. Its value is equal to the total number of angular nodes in the orbital and is indicated by the symbol 'l'.

The azimuthal quantum number can suggest an s, p, d, or f subshell, all of which have different forms. This value is determined by the principal quantum number (n). It has the values from zero to (n-1).

For example, if n = 3, the azimuthal quantum number can be 0, 1, or 2. The resulting subshell is 's' subshell when l = 0. When l = 1 and l = 2, the resulting subshells are 'p' and 'd' subshells, respectively. When n=3, the three subshells that can be used are 3s, 3p, and 3d.

In another example, if n = 5, the possible values of l are 0, 1, 2, 3, and 4.

3. Magnetic Quantum Number (m): (Lande)

The magnetic quantum number determines the subshell's overall number of orbitals and their orientation. The symbol ‘m’is used to represent it.

Because it gives the orientation or distribution of the electron cloud, it determines the direction of orientation of electrons in subshell.

The azimuthal (or orbital angular momentum) quantum number determines the value of the magnetic quantum number. The value of 'm' is dependent on the value of 'l'. It has all values from − l to + l including zero, with total of (2l + 1) values. It describes the splitting of spectral lines caused by a magnetic field (Zeeman effect).

4. Spin Quantum Number (s): (Uhlenbech and Goldschmidt)

It determines the orientation of the electron's spin. It has two values for electron spinning about its own axis. The spin quantum number either equal to – 1/2 (anti-clockwise) and + ½ (clockwise) totally depending on the spin of electron.

Pauli Exclusion Principle

This principle was given by Wolfgang Pauli (1926). This principle actually excludes the possibility of any two electrons having all the same four quantum numbers i.e., 'No two electrons in an atom can have the same set of four quantum numbers'. Because the n, l and m, quantum numbers address a particular orbital. The four quantum numbers required to define the spin of the electron orbital. It can have only two possible values i.e., $+\frac{1}{2}$ and $-\frac{1}{2}$. Therefore a maximum of only two electrons can be accommodated in an atomic orbital and these two electrons must have opposite spin.

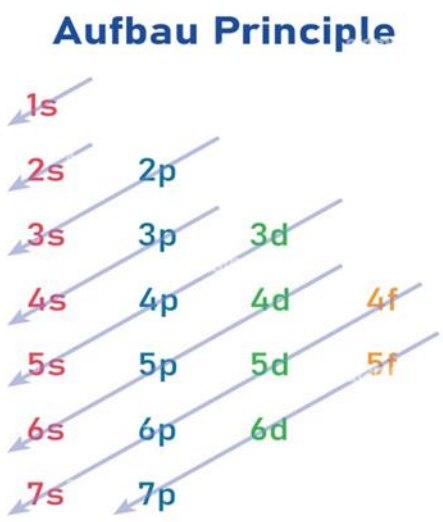
For example, helium has two electrons $1s^2$ that have the same n, l, m values but different values of spin quantum number.

Aufbau Principle

According to Aufbau Principle: "In a ground state of the atoms, the orbitals are filled in order of their increasing energies." i.e. an electron will initially occupy an orbital of lower energy level and when the lower energy level orbitals are occupied, then only they shall start occupying the higher energy level orbitals.

Salient Features of the Aufbau Principle

- Energy of an orbital is determined by (n+l) rule where 'n' stands for the Principal quantum number and 'l' stands for the Azimuthal quantum number. The lower the value of (n+l) for an orbital, lower will be its energy. And, if two orbitals have the same value for (n+l) then the one with higher value of n will have higher energy.
- Electron enters in that subshell which has minimum value of n + l and for equal value of n + l, electron goes in that subshell which has less vlue of n..
- The order in which the energies of the electronic orbitals increase and their respective order of filling as per the Aufbau rule is as follows:



Ex:
C (6) – $1s^2, 2s^2, 2p^2$
Na (11) - $1s^2, 2s^2, 2p^6, 3s^1$
Cl (17) - $1s^2, 2s^2, 2p^6, 3s^2, 3p^5$
Fe (26) - $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^6$

Hund's Rule

- This rule describes the order in which electrons are filled in all the orbitals belonging to a subshell.
- It states that every orbital in a given subshell is singly occupied by electrons before a second electron is filled in an orbital.
- In order to maximize the total spin, the electrons in the orbitals that only contain one electron all have the same spin (or the same values of the spin quantum number).

Electronic Configuration of Elements

The electronic configuration of an element is the arrangement of electrons in various shells, subshells, or orbitals of an atom. Electron configurations of atoms follow a standard notation in which all electron-containing atomic subshells (with the number of electrons they hold written in superscript) are placed in a sequence. For example, the electron configuration of sodium is $1s^2, 2s^2, 2p^6, 3s^1$.

However, the standard notation often yields lengthy electron configurations (especially for elements having a relatively large atomic number). In such cases, an abbreviated or condensed notation may be used instead of the standard notation. In the abbreviated notation, the sequence of completely filled subshells that correspond to the electronic configuration of

a noble gas is replaced with the symbol of that noble gas in square brackets. Therefore, the abbreviated electron configuration of sodium (Na) is-

Na (11) - [Ne] $3s^1$ (the electron configuration of neon is $1s^2, 2s^2, 2p^6$).

These three rules that dictate the manner in which electrons are filled in atomic orbitals are:

- **The Aufbau principle:** electrons must completely fill the atomic orbitals of a given energy level before occupying an orbital associated with a higher energy level. Electrons occupy orbitals in the increasing order of orbital energy level.
- **Pauli's exclusion principle:** states that no two electrons can have equal values for all four quantum numbers. Consequently, each subshell of an orbital can accommodate a maximum of 2 electrons and both these electrons must have opposite spins.
- **Hund's rule of maximum multiplicity:** All the subshells in an orbital must be singly occupied before any subshell is doubly occupied. Furthermore, the spin of all the electrons in the singly occupied subshells must be the same (in order to maximize the overall spin).

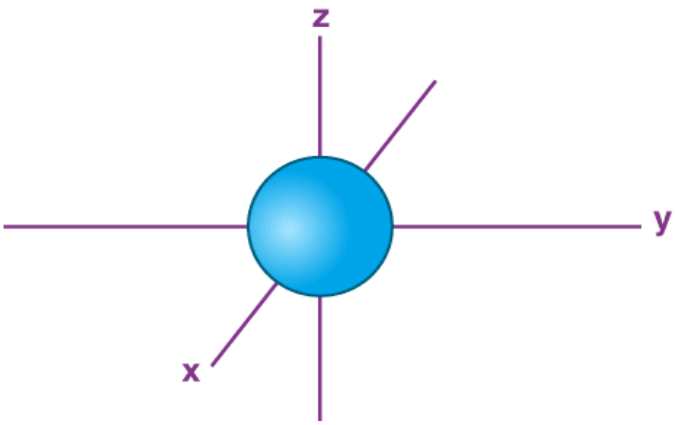
Shapes of orbitals

Atomic orbitals are mathematical functions that describe the wave nature of electrons (or electron pairs) in an atom. There are four different kinds of orbitals, denoted s, p, d and f each with a different shape. Of the four, s and p orbitals are considered because these orbitals are the most common in organic and biological chemistry.

An s-orbital is spherical with the nucleus at its centre, p-orbitals is dumbbell-shaped, d orbitals are double dumbbell shaped. The fifth d orbital is shaped like an elongated dumbbell with a doughnut around its middle. The orbitals in an atom are organized into different layers or electron shells.

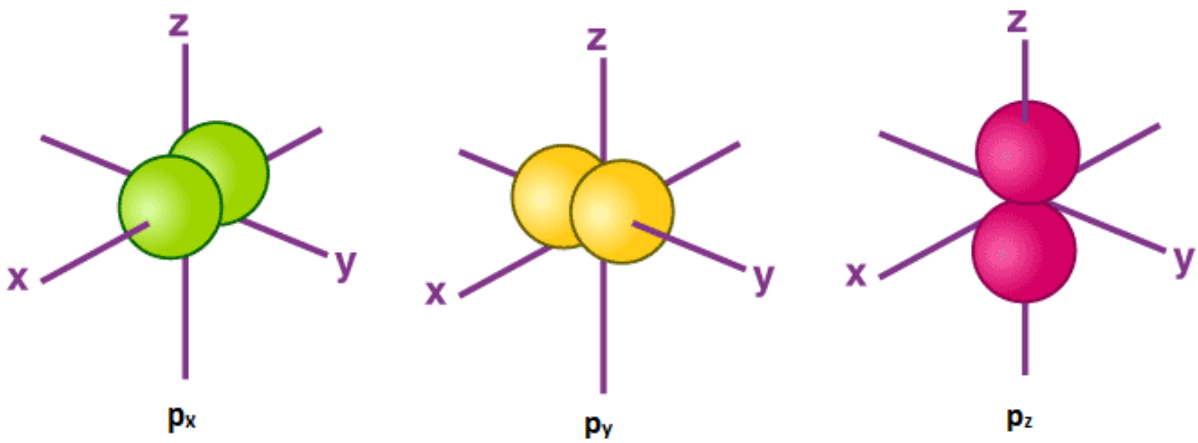
The Shape of s Orbitals

- The s orbital looks like a sphere having the nucleus as its centre which in two dimensions can be seen as a circle.
- Hence, we can say that s-orbitals are spherically symmetric having the probability of finding the electron at a given distance equal in all the directions.
- The size of the s orbital is also found to increase with the increase in the value of the principal quantum number (n), thus, **4s > 3s> 2s > 1s**.



The Shape of p Orbitals

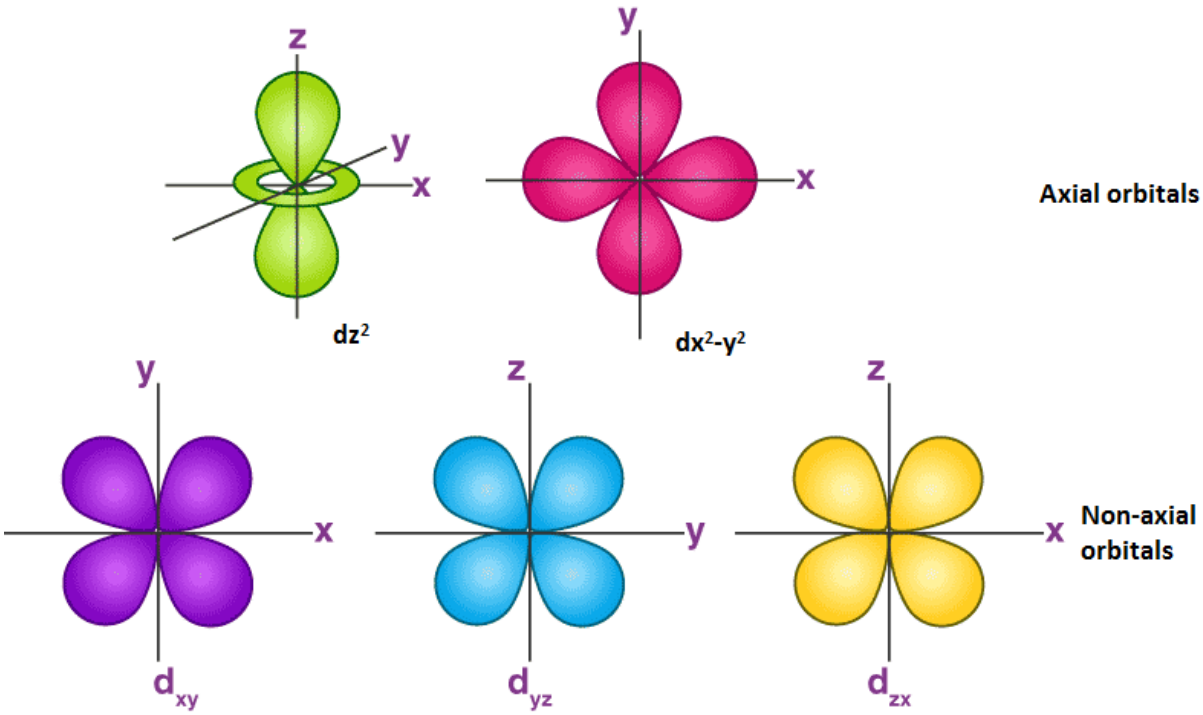
- Each p orbital consists of two sections better known as lobes which lie on either side of the plane passing through the nucleus.
- The three p orbitals differ in the way the lobes are oriented whereas they are identical in terms of size, shape, and energy.
- As the lobes lie along one of the x, y or z-axis, these three orbitals are given the designations $2p_x$, $2p_y$, and $2p_z$. Thus, we can say that there are three p orbitals whose axes are mutually perpendicular.
- Similar to s orbitals the size, and energy of p orbitals increase with an increase in the principal quantum number (4p > 3p > 2p).



The Shape of d Orbitals

- The magnetic orbital quantum number for d orbitals is given as (-2, -1, 0, 1, 2). Hence, we can say that there are five d-orbitals.
- These orbitals are designated as d_{xy}, d_{yz}, d_{xz}, $d_{x^2-y^2}$ and d_{z^2}.
- d_{xy}, d_{yz} and d_{xz} orbitals are non-axial whereas $d_{x^2-y^2}$ and d_{z^2} orbitals are axial.

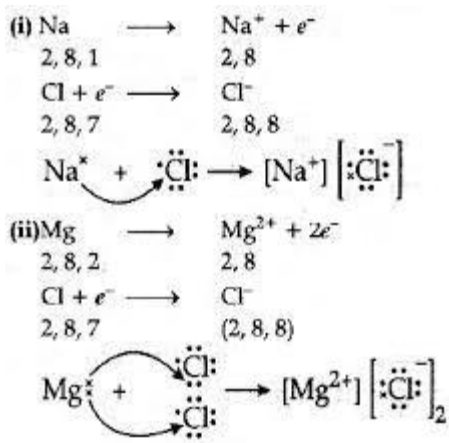
- Out of these five d orbitals, the shapes of the first four d-orbitals are similar to each other, which is different from the d_z^2 orbital whereas the energy of all five d orbitals is the same.



Ionic Bond:

The electrostatic interaction between positive and negative ions in a chemical molecule forms an ionic bond, also known as an electrovalent bond. A bond is created when the valence (outermost) electrons of one atom are transferred to outermost shell of another dissimilar atom. The atom which loses the electron acquires a positive charge while other atom which gains electrons acquires negative charge. It is non-directional in nature.

Ex: NaCl and $MgCl_2$



Factors affecting Ionic bond

- **Ionization energy:**

It is defined as the amount of energy required to remove an electron from an isolated gaseous atom of an element. The lesser the ionization energy, the greater is the ease of the formation of a cation.

$$M(g) + IE \rightarrow \underset{\text{Cation}}{M^+(g)} + e^-$$

- **Electron affinity:**

It is defined as the amount of energy released when an electron is added to an isolated gaseous atom of an element. The higher the energy released during this process, the easier will be the formation of an anion.

$$M(g) + e^- \rightarrow \underset{\text{Anion}}{M^-} + EA$$

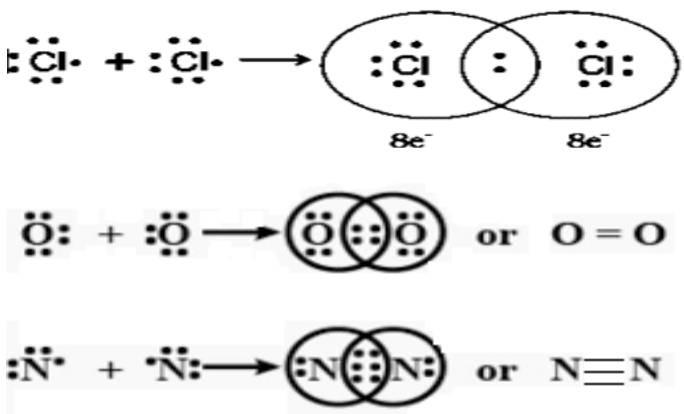
- **Lattice energy:**

It is defined as the amount of energy released when cations and anions are brought from infinity to their respective equilibrium sites in the crystal lattice to form one mole of the ionic compound. The higher the lattice energy, the greater is the tendency of the formation of an ionic bond. The higher the charges on the ions and smaller the distance between them, the greater is the force of attraction between them.

$$M^+(g) + X^-(g) \rightarrow \underset{\text{Ionic crystal}}{MX(s)} + U$$

Covalent bond

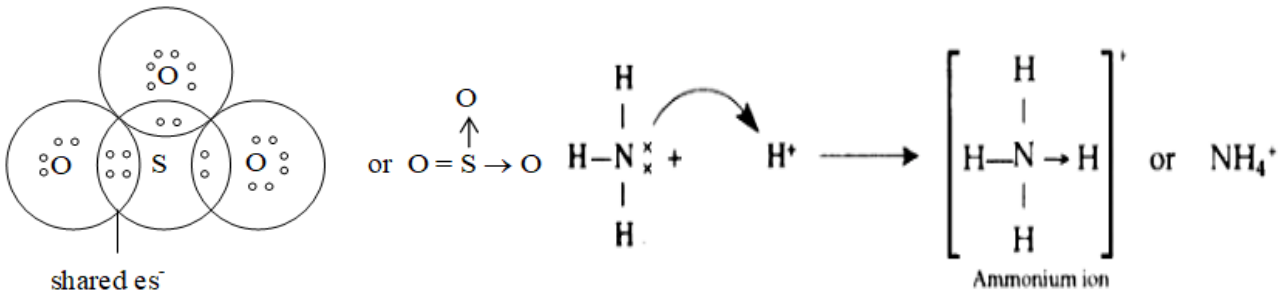
It involves sharing of equal electrons between two atoms. The number of electrons which an atom contributes for sharing in a covalent bond is called its covalency. It may be single (−), double (=) or triple (≡) depending upon number of electron pairs shared. It is directional in nature. Ex: Cl_2, O_2, N_2



Co-ordinate bond

The linkage, in which shared electrons are contributed by one atom, is called coordinate or dative bond. The atom which donates the pair of electron is called donor and other atom which accept pair of electrons, is called acceptor.

Ex: SO_3, NH_4^+



Hydrogen bonding

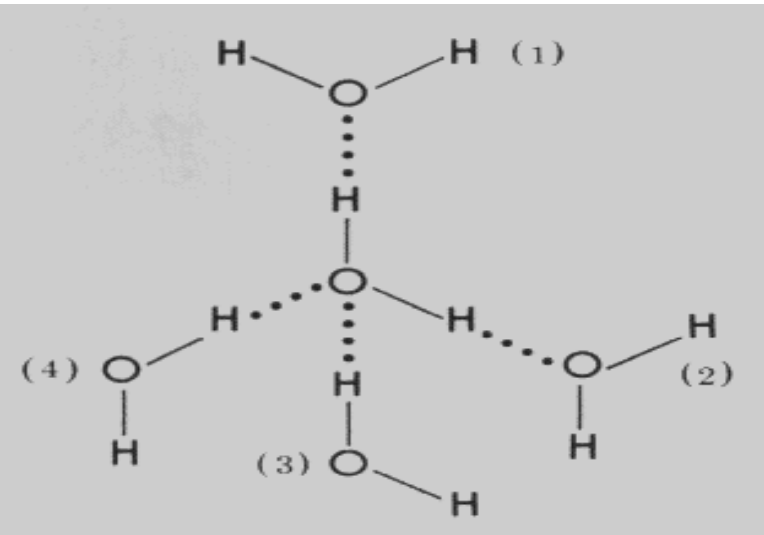
Hydrogen attached to electronegative atom (F, O, N) forms a bond with other electronegative atom (F, O, N). This bond is called hydrogen bond (H-bond). This is represented by dotted line. It is a weak bond.

Types of H-bonding: There are two types of H-bonding:

1. Intermolecular H-bonding
2. Intramolecular H-bonding

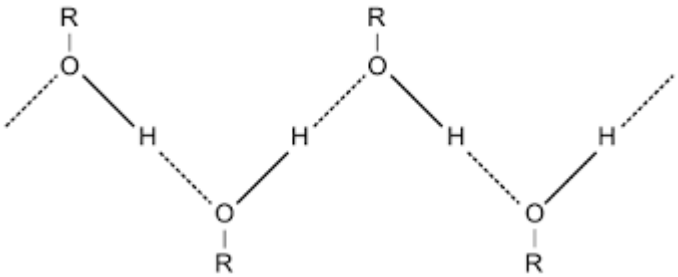
- **Intermolecular H-bonding:** In this type H-bonding takes place between H and electronegative element present in the different molecules of the same substance (in between H_2O and H_2O) or different substances (in between H_2O and ROH). Ex: H_2O, ROH, NH_3, HF, RCOOH etc.

In H_2O molecule:

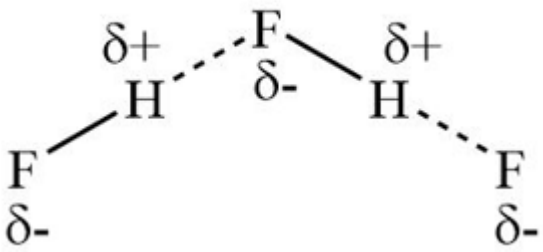


One H_2O molecule is attached to four other H_2O molecules- two with H-atoms and two with O-atoms. When ice is formed from water, some air is created in tetrahedral packing of water molecules. Due to this air gap, volume of ice is greater than water and thus ice is lighter than water. The density decreases when ice is formed. Water has maximum density at 4^0C.

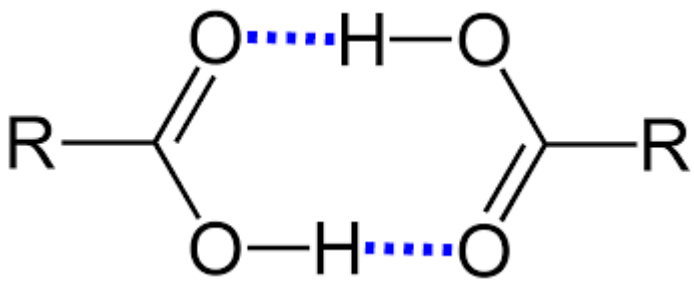
In ROH:



In HF:



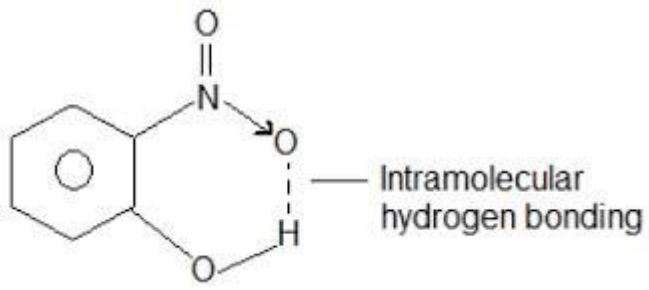
In Carboxylic acid: Carboxylic acid dimerises due to H-bonding.

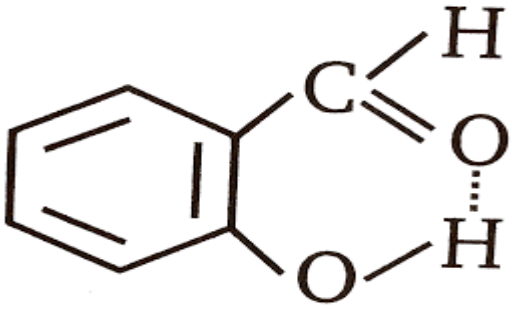


- **Intramolecular H-bonding:** This type of H-bonding occurs when polar hydrogen atom attached with electronegative atom present within same molecule.
  Conditions for the formation of Intramolecular H-bonding:
  i. A five or six membered ring should be formed.

ii. The ring form as a result of H-bonding should be planar. It is called chelation.
iii. Interacting atoms should be placed in such a way that there is minimum strain during the ring closure.

Ex: o-nitrophenol, o-hydroxybenzaldehyde etc.





Effect of Inter and Intramolecular H-bonding

- **Melting and boiling point:** Intermolecular H-bonding increases melting and boiling point.
- **Association of molecule:** Due to Intermolecular H-bonding two or more molecules of a compound get associated to form bigger units. Ex: in carboxylic acid dimer is formed even in vapour state.
- **Solubility:** Intermolecular H-bonding is responsible for solubility one compound in other compound. Ex: Alcohol (ROH) is soluble in water (H_2O).
- **Viscosity and surface tension:** Compounds which can form strong hydrogen bonds have higher values of viscosity and surface tension.
- **Volatility:** Intramolecular H-bonding increases volatility in compound. Ex: o-nitrophenol is more volatile than p-nitrophenol.

Valence Bond Theory

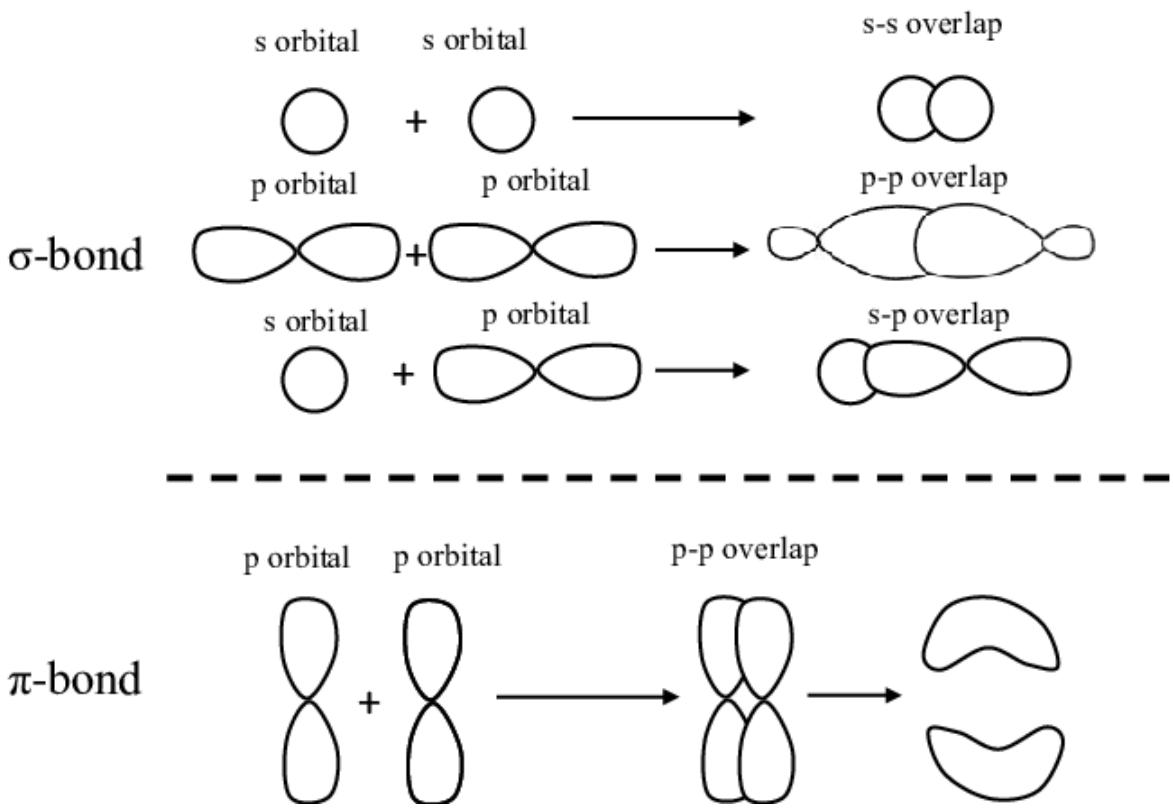
The valence bond theory was proposed by Heitler and London to explain the formation of covalent bond quantitatively using quantum mechanics. Later on, Linus Pauling improved this theory by introducing the concept of hybridization. The Valence Bond Theory was developed in order to explain chemical bonding using the method of quantum mechanics. This theory primarily focuses on the formation of individual bonds from the atomic orbitals of the participating atoms during the formation of a molecule.

Postulates of Valence Bond Theory

The postulates of the Valence Bond Theory are given below.

- The overlapping of two half-filled valence orbitals of two different atoms results in the formation of the covalent bond. The overlapping causes the electron density between two bonded atoms to increase. This gives the property of stability to the molecule, greater the extent of overlapping, stronger is the bond formed.

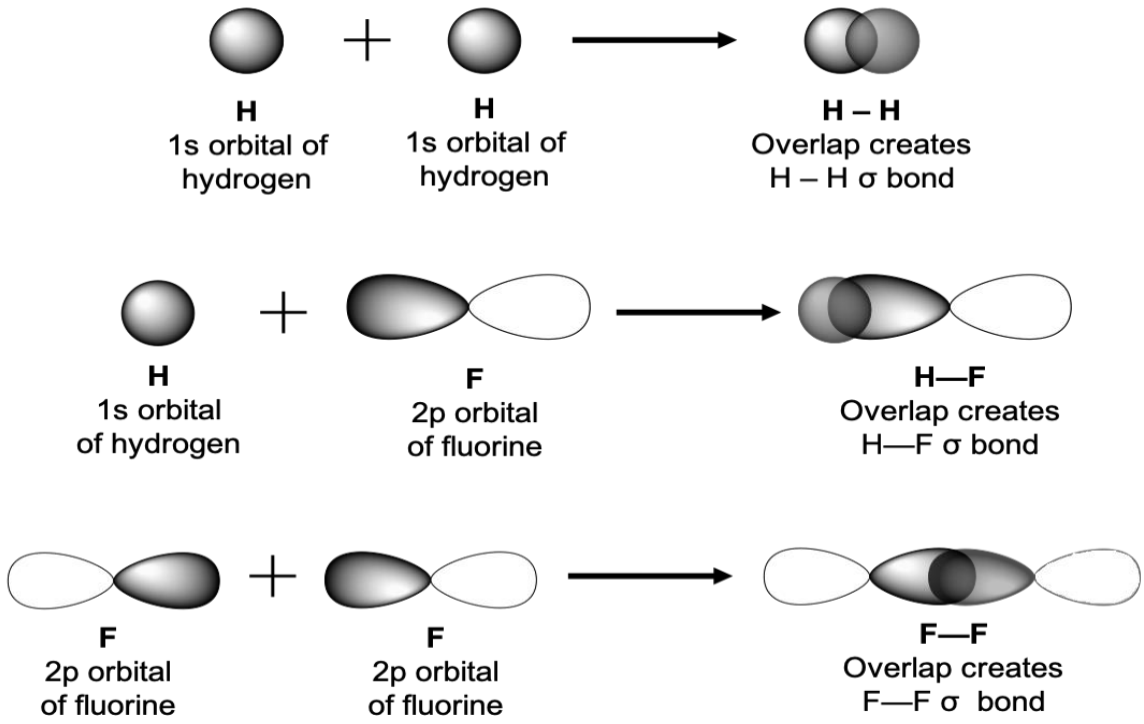
- In case the atomic orbitals possess more than one unpaired electron, more than one bond can be formed and electrons paired in the valence shell cannot take part in such a bond formation.
- The direction of the covalent bond is along the region of overlapping of the atomic orbitals, i.e. a covalent bond is directional.
- Based on the pattern of overlapping, there are two types of covalent bonds: sigma bond (σ-bond) and a pi bond (π-bond). The covalent bond produced by sideways overlapping of atomic orbitals is called pi bond whereas the bond created by overlapping of atomic orbital in between the inter nucleus axis is identified as a sigma bond.



This concept is used to describe the covalent bond creation in many molecules.

Ex: In F_2 molecule, the F−F bond is made by the intersection of pz orbitals of the two F atoms, each comprising an unpaired electron. Subsequently, the natures of the intersecting orbitals are different in H_2 hydrogen and F_2 fluorine molecules, the bond strength and bond lengths differ between H_2 and F_2 molecules.

Ex: In an HF molecule, the covalent bond is made by the intersection of the 1s orbital of H hydrogen and the 2pz orbital of F fluorine, each comprising an unpaired electron. Common sharing of electrons between H and F results in a covalent bond in HF.



Molecular Orbital Theory

The molecular orbital theory (MOT) is a theory on chemical bonding developed by F. Hund and R. S. Mulliken to describe the structure and properties of different molecules. The valence-bond theory failed to explain how certain molecules contain two or more equivalent bonds whose bond orders lie between that of a single bond and that of a double bond, such as the bonds in resonance-stabilised molecules. This is where the molecular orbital theory proved to be more powerful than the valence-bond theory.

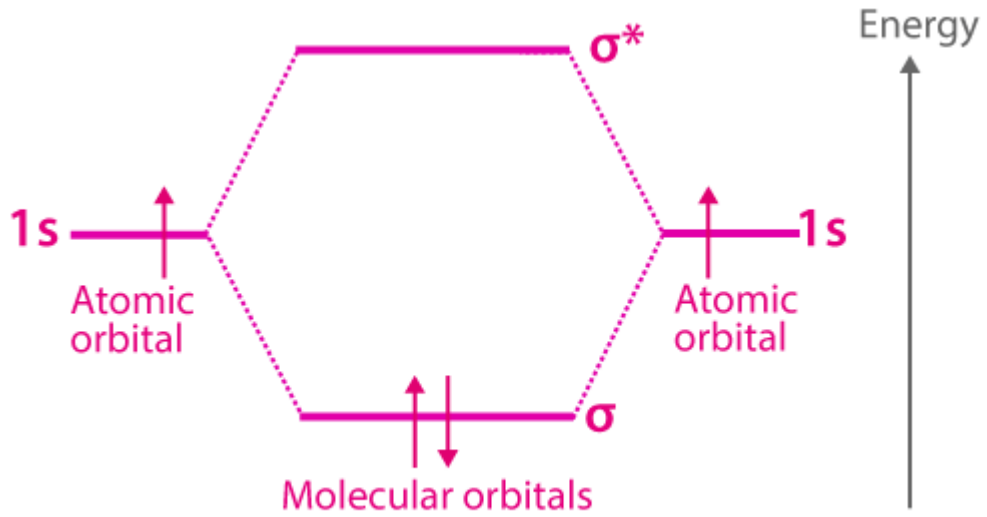
Molecular orbital theory approximation of the molecular orbitals as linear combinations of atomic orbitals (LCAO) can be illustrated as follows.

- The atomic orbitals overlap to form new orbitals called molecular orbitals. When two atomic orbitals overlap, they lose their identity and form new orbitals called molecular orbitals.
- The electrons in the molecules are filled in the new energy states called the molecular orbitals, similar to the electrons in an atom being filled in an energy state called atomic orbitals.
- The probability of finding the electronic distribution in a molecule around its group of nuclei is given by the molecular orbital.
- The two combining atomic orbitals should possess energies of comparable value and similar orientation. For example, 1s can combine with 1s and not with 2s.
- The number of molecular orbitals formed is equal to the number of atomic orbitals combined.

- The shape of molecular orbitals formed depends upon the shape of the combining atomic orbitals.

According to the molecular orbital theory, the filling of orbitals takes place according to the following rules:

- **Aufbau's principle:** Molecular orbitals are filled in the increasing order of energy levels.
- **Pauli's exclusion principle:** In an atom or a molecule, no two electrons can have the same set of four quantum numbers.
- **Hund's rule of maximum multiplicity:** Pairing of electrons doesn't take place until all the atomic or molecular orbitals are singly occupied.



Molecular Orbital Diagram of H_2 molecule

VSEPR theory

The VSEPR model based largely on Lewis structures provides relatively simple and straight forward methods for predicting the geometry of molecules. This theory is used for predicting the shapes of covalent compounds. The repulsive interactions of electron pairs in the valence shell of the atoms are the basis for this theory. VSEPR is abbreviated as valence shell electron pair repulsion. According to this Theory:

- The shape of a molecule depends upon the number of valence shell electron pairs (bonded or non-bonded) around the central atom.
- Pairs of electrons in the valence shell repel one another since their electron clouds are negatively charged.
- These pairs of electrons tend to occupy such positions in space that minimise repulsion and thus maximise the distance between them.
- The valence shell is taken as a sphere with the electron pairs localising on the spherical surface at maximum distance from one another.
- If central atom is surrounded only by hybrid orbitals containing shared pairs of electrons (bond pairs), molecule has regular geometry. If hybrid orbitals contain shared pairs of electrons as well as unshared pairs of electrons (lone pairs), molecule has distorted geometry.

- Where two or more resonance structures can represent a molecule, the VSEPR model is applicable to any such structure.

  Electron pair's repulsive interactions decrease in the following order:
  Lone pair (lp) – Lone pair (lp) > Lone pair (lp) – Bond pair (bp) > Bond pair (bp) – Bond pair (bp)

Hybridisation

In Chemistry, hybridisation is the combination of two or more atomic orbitals with different energies to form new atomic orbitals with equal energy. Fully filled and half-filled orbitals, however, can both participate in this process if their energies are equivalent. The term "hybridisation" was first used by British chemist Frederick Crick.

Characteristics of Hybridization

Some of the following characteristics of Hybridization is give below-:

1. Hybridization occurs when orbitals belonging to the same atom or ion have similar energies.

2. The number of hybrid orbitals equals the number of orbitals involved in the hybridization process.

3. In terms of energy and shape, hybrid orbitals are always comparable.

4. In comparison to pure atom orbitals, hybrid orbitals generate more stable bonds.

5. The goal of hybridization is to create equivalent orbitals that have the most symmetry.

6. It is unknown whether or not hybridization actually occurs. It's a theory that describes how molecules behave in the real world.

7. The hybrid orbitals are directed in space in the same desired direction in order to have a stable arrangement and provide the molecule a proper shape.

Types of Hybridization

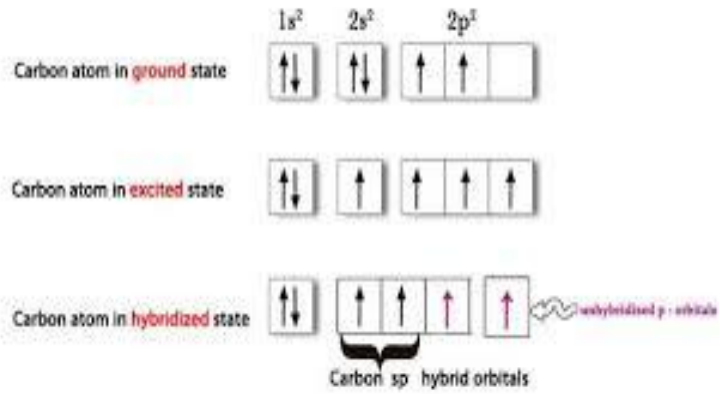
The types of hybridization are as follows:

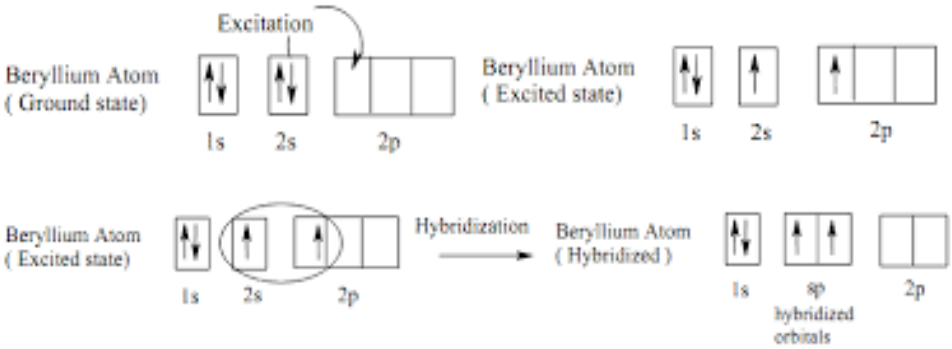
1) sp Hybridisation

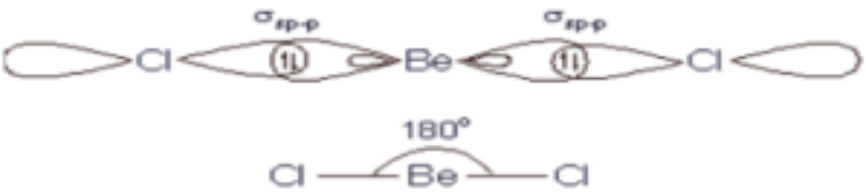
In sp hybridization one s- and one p-orbital are combined together to generate two sp – hybrid orbitals with a linear structure and a bond angle of 180 degrees.

Each sp hybridised orbital has the same proportion of s and p characters – 50 % s character and 50 % p character.

For example, when $BeCl_2$ is formed, the first atom is in the excited state $2s^1 2p^1$, which is then hybridised to generate two sp – hybrid orbitals. $BeCl_2$ is formed when these hybrid orbitals collide with the two p-orbitals of two chlorine atoms. Other example is acetylene.





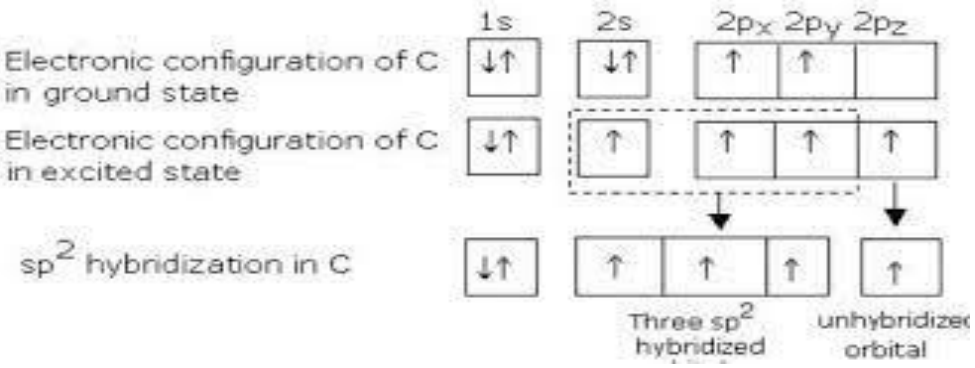


2) sp^2 Hybridisation

One s- and one p-orbital are combined together to generate three sp2– hybrid orbitals with a triangular planar shape and a bond angle of 120 degrees. These three hybrid orbitals remain in the same plane and form a 120° angle with one another.

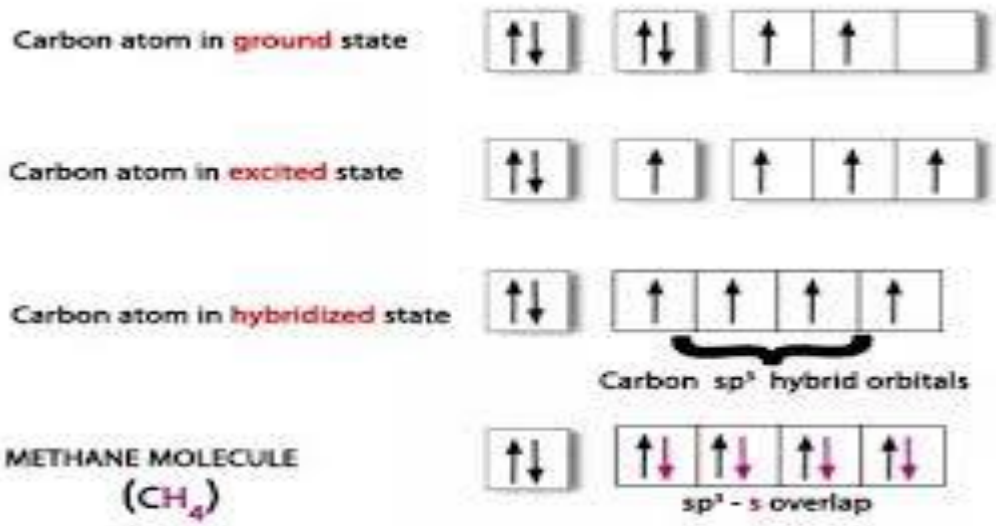
The hybrid orbitals have a 33.3 % 's' character and a 66.66 % 'p' character in each of them.

Ex: Ethylene, BCl_3 etc.



3) **sp^3 Hybridisation**

One s- and three p- orbitals are combined together to generate four sp^3– hybrid orbitals with a tetrahedral structure. These are aimed at the four corners of a conventional tetrahedron and form a 109°28′ angle with each other.

Each sp^3 hybrid orbital contains 25% s character and 75% p character.

Ex: Ethane, Methane etc.